



流体力学方程组学术研讨会



会议时间：2026 年 1 月 16 日-19 日

主办单位：北京大学数学科学学院



北京大学数学科学学院
School of Mathematical Sciences
Peking University

流体力学方程组学术研讨会

会议通知

尊敬的_____教授

您好！

为了促进偏微分方程领域内不同方向的交叉融合，特别是加强流体力学方程组与椭圆方程方向之间的学术交流与合作，兹定于 2026 年 1 月 16 日至 1 月 19 日在北京大学举办“流体力学方程组学术研讨会”。会议将重点围绕流体力学方程组及椭圆方程理论的最新研究成果、关键科学问题与未来发展趋势展开深入研讨，推动相关理论方法与实际应用的创新发展。衷心期待各位专家学者莅临指导。

一、学术委员会：

张平（中科院数学院） 章志飞（北京大学）

二、组织委员会：

欧阳智萌（北京大学） 向圣权（北京大学） 韦东奕（北京大学） 王超（北京大学）

三、会议时间与地点：

报到注册：1 月 16 日

报告时间：1 月 17 日、1 月 18 日

离会时间：1 月 19 日

会议地点：北京大学智华楼王选报告厅

四、联系人：

王 超： 15810330497 wangchao@math.pku.edu.cn

张欣颖： 15901466403 zhangxy@math.pku.edu.cn

五、本次会议不收取会议注册费，住宿统一安排，参会嘉宾住宿费、餐饮轨迹交通费请自理。

此致

敬礼！

北京大学数学科学学院

2026 年 1 月 8 日

会议日程:

2026-01-17

时间	报告题目	报告人	主持人
8:40-9:00	开幕致辞、大合影		
9:00-9:40	Uniqueness and stability of the steady vortex ring of 3D incompressible Euler equation	曹道民	李海梁
9:45-10:25	Singular limits of the compressible ideal MHD equations	琚强昌	郭玉劲
10:25-10:55	茶歇		
10:55-11:35	Global Well-Posedness for the 2D and 3D Prandtl-Shercliff Model	李维喜	王术
12:00-14:00	农园三楼		
14:00-14:40	Kinetic Equations with External Potentials: Equilibria and Stability	何凌冰	孙永忠
14:45-15:25	$\mathcal{I}^2$ -decoupling and the unconditional uniqueness for the Boltzmann equation	沈舜麟	桂贵龙
15:25-15:55	茶歇		
15:55-16:35	Decay character theory for partially dissipative hyperbolic systems of balance laws	徐江	王伟
16:40-17:20	Landau-de Gennes model with sextic potentials: asymptotic behavior of minimizers	王尉	刘祖汉
18:00	晚餐		

2026-01-18

时间	报告题目	报告人	主持人
9:00-9:40	Spectral approach to the Boltzmann equation in a 3D infinite layer	刘双乾	章志飞
9:45-10:25	Asymptotic linear stability of a class of columnar flow	缪爽	韦东奕
10:25-10:55	茶歇		
10:55-11:35	Unstable mode around the 3D boundary layer flow	张筑	王超
12:00-14:00	农园三楼		
14:00-17:00	自由讨论		
18:00	晚餐		

报告信息:

Title: Uniqueness and stability of the steady vortex ring of 3D incompressible Euler equation

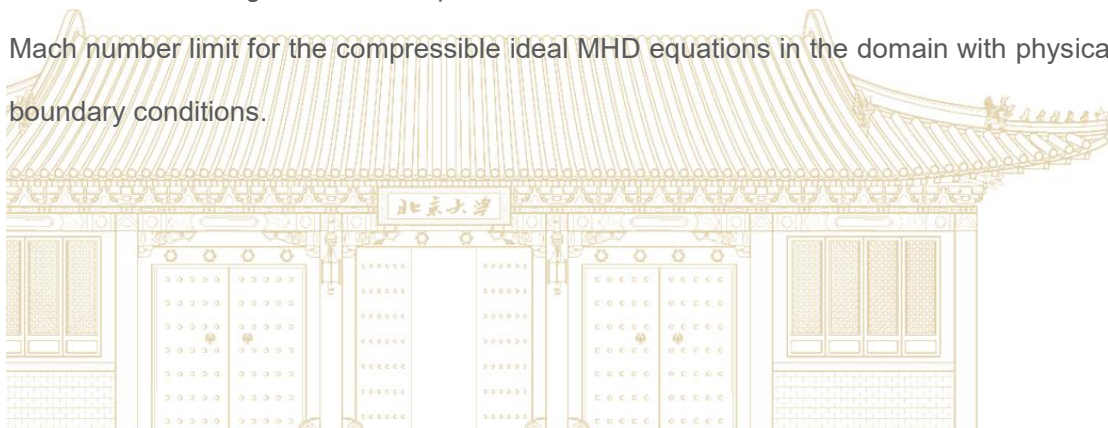
Speaker: 曹道民 (中国科学院数学与系统科学研究院)

Abstract: For the 2D incompressible Euler equation, there is a lot of results on the existence of vortex solutions, such as solutions with vorticity compactly supported or vorticity concentrated in several small sets. For the 3D case, the situation is much more difficult. A well-known solution is the vortex ring with small cross-section, traveling with a constant speed along its symmetric-axis, which was observed by Helmholtz in 1858. To get existence of solutions, people usually consider axi-symmetric case and a lot of axi-symmetric solutions have been established by different methods. But very little is known about the uniqueness. The stability of such vortex ring with small cross-section is a long-time open problem. In this talk the speaker will first talk about the uniqueness of such vortex ring. Then the speaker will introduce results on the nonlinear stability of such vortex ring. The results presented in this talk are from joint papers with Guolin Qin, Weilin Yu, Weicheng Zhan and Changjun Zou.

Title: Singular limits of the compressible ideal MHD equations

Speaker: 琚强昌 (北京应用物理与计算数学研究所)

Abstract: In this talk, I will review the main results of singular limits of the compressible ideal MHD equations including the low Mach number limit, small Alfvén number limit and the three-scale singular limit. In particular, I will introduce the recent work on the low Mach number limit for the compressible ideal MHD equations in the domain with physical boundary conditions.



Title: Global Well-Posedness for the 2D and 3D Prandtl-Shercliff Model

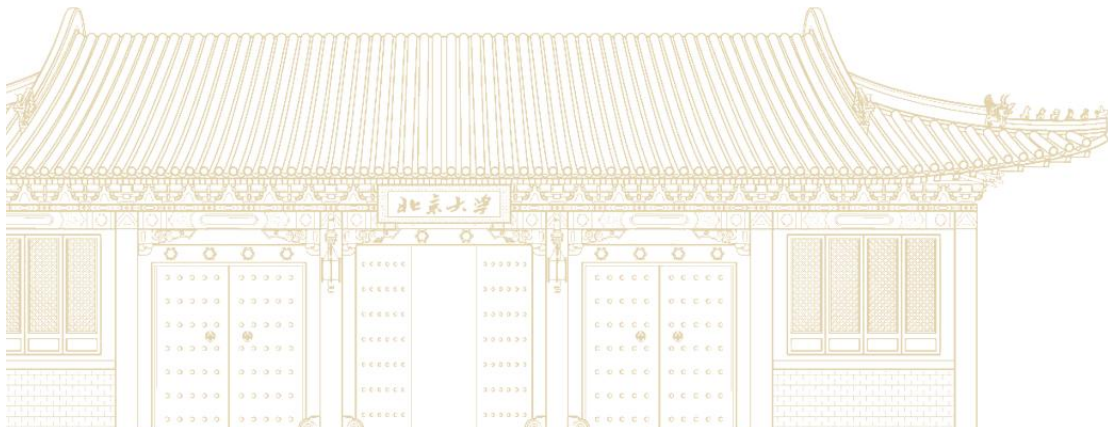
Speaker: 李维喜 (武汉大学)

Abstract: We investigate the Prandtl-Shercliff model in both two and three dimensions. For the two-dimensional case, we establish global-in-time well-posedness in Sobolev spaces without any structural assumptions on the initial data. Furthermore, we show that the solution exhibits an analytic regularization effect in all variables, which holds globally in time and in space up to the boundary. For the three-dimensional case, we study a linearized version of the model and prove its global-in-time well-posedness for initial data that are analytic in only one tangential direction. The proofs rely crucially on the intrinsic non-local diffusion induced by the Shercliff boundary layer.

Title: Kinetic Equations with External Potentials: Equilibria and Stability

Speaker: 何凌冰 (清华大学)

Abstract: We study the Boltzmann and Landau equations in the presence of external potentials. Our main results establish that: (i) Every entropy-invariant solution is uniquely determined by the conservation laws. (ii) The equilibria are linearly or nonlinearly stable, with polynomial convergence rates that are explicitly characterized by the initial data.



Title: l^2 -decoupling and the unconditional uniqueness for the Boltzmann equation

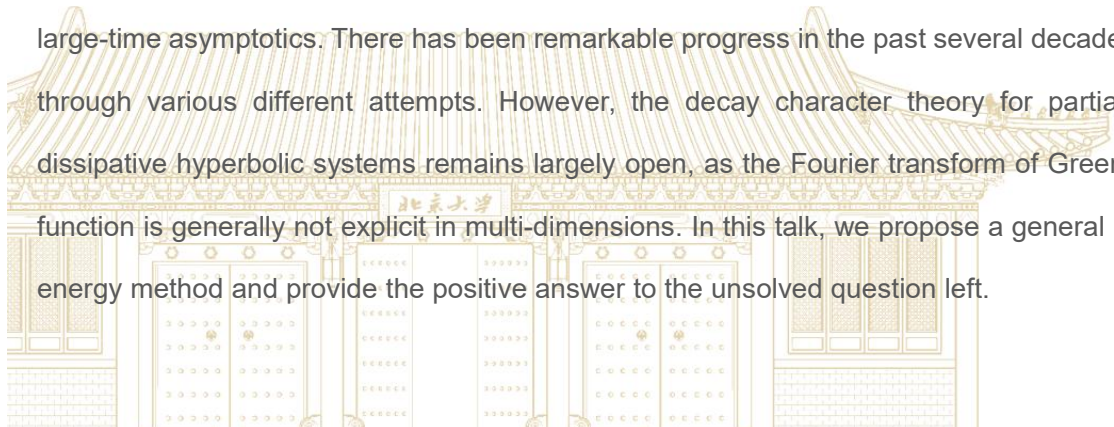
Speaker: 沈舜麟 (中国科学技术大学)

Abstract: We broaden the application of the l^2 -decoupling theorem to the Boltzmann equation. We prove full-range Strichartz estimates for the linear problem in the $\mathbb{T}^d$ setting. We establish space-time bilinear estimates, and hence the unconditional uniqueness of solutions to the $\mathbb{R}^d$ and $\mathbb{T}^d$ Boltzmann equation for the Maxwellian particle and soft potential with an angular cutoff, adopting a unified hierarchy scheme originally developed for the nonlinear Schrödinger equation. This talk is based on a joint work with Xuwen Chen and Zhifei Zhang.

Title: Decay character theory for partially dissipative hyperbolic systems of balance laws

Speaker: 徐江 (南京航空航天大学)

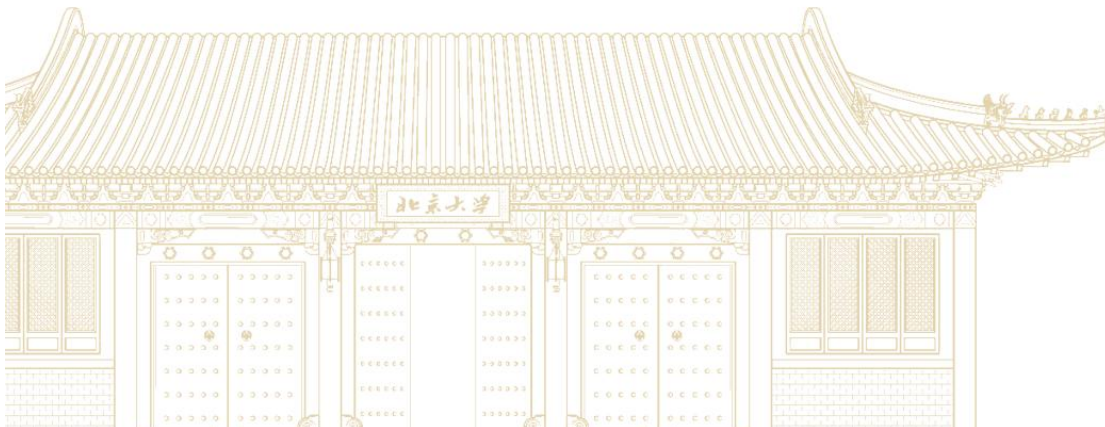
Abstract: The partially dissipative systems that characterize many physical phenomena were first pointed out by Godunov (1961), then investigated by Friedrichs-Lax (1971) who introduced the convex entropy, and later by Shizuta-Kawashima (1984,1985) who initiated a simple sufficient criterion ensuring the global existence of smooth solutions and their large-time asymptotics. There has been remarkable progress in the past several decades, through various different attempts. However, the decay character theory for partially dissipative hyperbolic systems remains largely open, as the Fourier transform of Green's function is generally not explicit in multi-dimensions. In this talk, we propose a general L_p energy method and provide the positive answer to the unsolved question left.



Title: Landau-de Gennes model with sextic potentials: asymptotic behavior of minimizers

Speaker: 王尉 (北京大学)

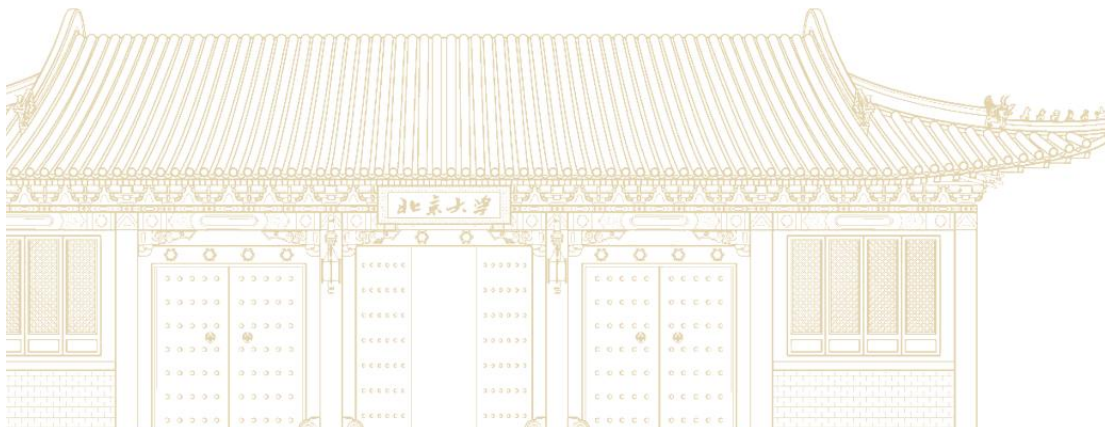
Abstract: We study a class of Landau-de Gennes energy functionals with sextic bulk energy densities in a three-dimensional domain. We examine the asymptotic behavior of minimizers in two distinct scenarios: one in which their energy remains uniformly bounded, and another in which the energy logarithmically diverges as the elastic constant tends to zero. In the first case, we show that, up to a subsequence, the minimizers converge to a locally energy-minimizing harmonic map in the H_{loc}^1 -topology. Moreover, on every compact subset that is away from the singularities of the limiting map, we prove convergence in C_{loc}^j for any $j \in \mathbb{Z}_+$. In the second case, we identify a closed set of line defects, denoted by S_{line} . It has finite length and consists of finite segments of lines locally, such that the energies of minimizers are locally uniformly bounded away from it. This work solves an open question raised by Canevari (Arch. Ration. Mech. Anal. 223 (2017), 591-676), specifically concerning point and line defects in the Landau-de Gennes model with sextic potentials.



Title: Spectral approach to the Boltzmann equation in a 3D infinite layer

Speaker: 刘双乾 (华中师范大学)

Abstract: In this talk, I will present a spectral approach to the study of the large-time asymptotic behavior of solutions to the Boltzmann equation near global Maxwellians in a three-dimensional infinite layer $\mathbb{R}^2 \times (-1,1)$. The isothermal diffuse reflection boundary condition is imposed on two parallel infinite planes at $x_3 = \pm 1$. The main difficulties lie in the fact that the direct Fourier transform is not applicable to the vertical x_3 -variable, and the linear collision operator K loses its compactness on $L^2((-1,1) \times \mathbb{R}^3_v)$ although it is compact on $L^2(\mathbb{R}^3_v)$. By introducing a regularization operator K_n via the finite-dimensional Fourier series truncation in $L^2(-1,1)$, we study the spectrum of the linearized initial-boundary value approximation problem, establish the resolvent estimates, and identify the leading diffusive eigenvalue. This spectral structure governs the sharp asymptotic dynamics of the original linear problem as $n \rightarrow \infty$, enabling us to construct the large-time behavior for the nonlinear problem and rigorously prove that the solution converges with a faster rate toward that of the two-dimensional heat equation in the horizontal direction.



Title: Asymptotic linear stability of a class of columnar flow

Speaker: 缪爽 (武汉大学)

Abstract: Constructing dynamical global-in-time solutions to 3D incompressible Euler equation is in general challenging. Recently Guo-Pausader-Widmayer made a breakthrough in this direction by proving asymptotic stability of a uniform-rotating static solution. Their approach heavily depends on the “uniform-rotating” of the background solution. In this talk I will present our recent progress on asymptotic linear stability of a more general columnar flow with non-uniform rotation. This is a joint work with Siqi Ren and Zhifei Zhang.

Title: Unstable mode around the 3D boundary layer flow

Speaker: 张筑 (香港理工大学)

Abstract: We study the stability properties of boundary layer-type shear flows for the three-dimensional Navier-Stokes equations in the limit of small viscosity $0 < \nu \ll 1$. When the streamwise and spanwise velocity profiles are linearly independent near the boundary, we construct a crossflow mode that exhibits rapid growth at the rate of $e^{t/\sqrt{\nu}}$. Our results reveal an analytic instability in the three-dimensional Navier-Stokes equations around generic boundary layer profiles.

This instability arises from the interplay between spanwise flow and three-dimensional perturbations, and does not occur in purely two-dimensional flows.

